Sparse sets in hypergraphs

Wojciech Samotij
Tel Aviv University

Joint work with Gady Kozma, Asaf Shapira, and Adam Zsolt Wagner

Combinatorics at the Confluence
Pittsburgh, July 20–22

General setting

Let $\mathcal{H}$ be an r-uniform hypergraph with vertex set V and $N := |V|$.

Induced subhypergraph. For $W \subseteq V$, write

$$\mathcal{H}[W] := \{A \in \mathcal{H} : A \subseteq W\}.$$

Independent set. A set $I \subseteq V$ is independent in $\mathcal{H}$ if

$$\mathcal{H}[I] = \varnothing.$$

We denote

$$\mathcal{I}(\mathcal{H}) := \{I \subseteq V : I \text{ is independent in } \mathcal{H}\}, \qquad \alpha(\mathcal{H}) := \max_{I \in \mathcal{I}(\mathcal{H})} |I|.$$

General setting

Let $\mathcal{H}$ be an r-uniform hypergraph with vertex set V and $N := |V|$.

Induced subhypergraph. For $W \subseteq V$, write

$$\mathcal{H}[W] := \{A \in \mathcal{H} : A \subseteq W\}.$$

Independent set. A set $I \subseteq V$ is independent in $\mathcal{H}$ if

$$\mathcal{H}[I] = \varnothing.$$

We denote

$$\mathcal{I}(\mathcal{H}) := \{I \subseteq V : I \text{ is independent in } \mathcal{H}\}, \qquad \alpha(\mathcal{H}) := \max_{I \in \mathcal{I}(\mathcal{H})} |I|.$$

Independent sets $\longleftrightarrow$ **avoiding patterns**.

Motivating example

Let $\mathcal{T}_n$ be the 3-uniform hypergraph of triangles in K_n:

$$V(\mathcal{T}_n) = E(K_n), \qquad \mathcal{T}_n = \big\{\{ij, ik, jk\} : 1 \leqslant i < j < k \leqslant n\big\}.$$

Thus a graph $G \subseteq E(K_n)$ is triangle-free iff $G \in \mathcal{I}(\mathcal{T}_n)$.

Motivating example

Let $\mathcal{T}_n$ be the 3-uniform hypergraph of triangles in K_n:

$$V(\mathcal{T}_n) = E(K_n), \qquad \mathcal{T}_n = \big\{\{ij, ik, jk\} : 1 \leqslant i < j < k \leqslant n\big\}.$$

Thus a graph $G \subseteq E(K_n)$ is triangle-free iff $G \in \mathcal{I}(\mathcal{T}_n)$.

Theorem (Mantel, 1907)

$$\alpha(\mathcal{T}_n) = \mathrm{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor = \left(\frac{1}{2} + o(1)\right) \cdot \binom{n}{2}.$$

Motivating example

Let $\mathcal{T}_n$ be the 3-uniform hypergraph of triangles in K_n:

$$V(\mathcal{T}_n) = E(K_n), \qquad \mathcal{T}_n = \big\{\{ij, ik, jk\} : 1 \leqslant i < j < k \leqslant n\big\}.$$

Thus a graph $G \subseteq E(K_n)$ is triangle-free iff $G \in \mathcal{I}(\mathcal{T}_n)$.

Theorem (Mantel, 1907)

$$\alpha(\mathcal{T}_n) = \mathrm{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor = \left(\frac{1}{2} + o(1)\right) \cdot \binom{n}{2}.$$

Theorem (Ramsey, 1930)

For every $r \geqslant 2$ and all $n \geqslant n(r)$, there are no $G_1, \ldots, G_r \in \mathcal{I}(\mathcal{T}_n)$ s.t.

$$E(K_n) = G_1 \cup \cdots \cup G_r.$$

Random analogues

Question

Under what assumptions on p does $G \sim G_{n,p}$ satisfy, a.a.s.?

Mantel's theorem in $G_{n,p}$.

$$\alpha(\mathcal{T}_n[G]) = \left(\frac{1}{2} + o(1)\right) \binom{n}{2} p.$$

Ramsey's theorem in $G_{n,p}$. There are no graphs $G_1, \ldots, G_r \in \mathcal{I}(\mathcal{T}_n[G])$ such that

$$G = G_1 \cup \cdots \cup G_r.$$

Random analogues

Question

Under what assumptions on p does $G \sim G_{n,p}$ satisfy, a.a.s.?

Mantel's theorem in $G_{n,p}$.

$$\alpha(\mathcal{T}_n[G]) = \left(\frac{1}{2} + o(1)\right)\binom{n}{2}p.$$

Ramsey's theorem in $G_{n,p}$. There are no graphs $G_1, \ldots, G_r \in \mathcal{I}(\mathcal{T}_n[G])$ such that

$$G = G_1 \cup \cdots \cup G_r.$$

Answer: $p \gg n^{-1/2}$.

Frank–Rödl, Haxell–Kohayakawa–Łuczak, Kohayakawa–Łuczak–Rödl, Łuczak–Ruciński–Voigt, Rödl–Ruciński

Two general tools

Around 2010, the following two general theorems were proved.

There is a threshold function $p_{\mathcal{H}}$ such that, when $p \gg p_{\mathcal{H}}$:

Two general tools

Around 2010, the following two general theorems were proved.

There is a threshold function $p_{\mathcal{H}}$ such that, when $p \gg p_{\mathcal{H}}$:

Transference principle A (Schacht). With very high probability,

$$\alpha(\mathcal{H}[V_p]) = p\,\alpha(\mathcal{H}) \pm o(pN).$$

Two general tools

Around 2010, the following two general theorems were proved.

There is a threshold function $p_{\mathcal{H}}$ such that, when $p \gg p_{\mathcal{H}}$:

Transference principle A (Schacht). With very high probability,

$$\alpha(\mathcal{H}[V_p]) = p\,\alpha(\mathcal{H}) \pm o(pN).$$

Transference principle B (Conlon–Gowers). With high probability, every $R \subseteq V_p$ has a model $W \subseteq V$ satisfying

$$|R| = p \cdot \big(|W| \pm o(N)\big), \qquad |\mathcal{H}[R]| = p^r \cdot \big(|\mathcal{H}[W]| \pm o(|\mathcal{H}|)\big).$$

Transference B $\Rightarrow$ Transference A

Transference B gives, for each $R \subseteq V_p$, a model $W \subseteq V$ such that

$$|R| = p \cdot \big(|W| \pm o(N)\big), \qquad |\mathcal{H}[R]| = p^r \cdot \big(|\mathcal{H}[W]| \pm o(|\mathcal{H}|)\big).$$

Transference B $\Rightarrow$ Transference A

Transference B gives, for each $R \subseteq V_p$, a model $W \subseteq V$ such that

$$|R| = p \cdot \big(|W| \pm o(N)\big), \qquad |\mathcal{H}[R]| = p^r \cdot \big(|\mathcal{H}[W]| \pm o(|\mathcal{H}|)\big).$$

Upper bound. If $R \in \mathcal{I}(\mathcal{H}[V_p])$, then $|\mathcal{H}[R]| = 0$, and hence

$$|\mathcal{H}[W]| = o(|\mathcal{H}|).$$

Therefore, $|W| \leqslant \alpha(\mathcal{H}) + o(N)$ and

$$|R| \leqslant p\,\alpha(\mathcal{H}) + o(pN).$$

Transference B $\Rightarrow$ Transference A

Transference B gives, for each $R \subseteq V_p$, a model $W \subseteq V$ such that

$$|R| = p \cdot \big(|W| \pm o(N)\big), \qquad |\mathcal{H}[R]| = p^r \cdot \big(|\mathcal{H}[W]| \pm o(|\mathcal{H}|)\big).$$

Upper bound. If $R \in \mathcal{I}(\mathcal{H}[V_p])$, then $|\mathcal{H}[R]| = 0$, and hence

$$|\mathcal{H}[W]| = o(|\mathcal{H}|).$$

Therefore, $|W| \leqslant \alpha(\mathcal{H}) + o(N)$ and

$$|R| \leqslant p\,\alpha(\mathcal{H}) + o(pN).$$

Lower bound. Consider $V_p \cap I$ for a largest independent set $I \in \mathcal{I}(\mathcal{H})$. Standard tail bounds give $|V_p \cap I| = p\alpha(\mathcal{H}) + o(pN)$.

Container method

Container Lemma (Balogh–Morris–S.; Saxton–Thomason).
There is a family $\mathcal{C} \subseteq 2^V$ with

$$\log |\mathcal{C}| \leqslant O\big(p_{\mathcal{H}} N \log(1/p_{\mathcal{H}})\big)$$

such that

- every $I \in \mathcal{I}(\mathcal{H})$ is contained in some $C \in \mathcal{C}$;
- every $C \in \mathcal{C}$ is almost independent in $\mathcal{H}$: $|\mathcal{H}[C]| = o(|\mathcal{H}|)$.

Container method

Container Lemma (Balogh–Morris–S.; Saxton–Thomason). There is a family $\mathcal{C} \subseteq 2^V$ with

$$\log |\mathcal{C}| \leqslant O\big(p_{\mathcal{H}} N \log(1/p_{\mathcal{H}})\big)$$

such that

- every $I \in \mathcal{I}(\mathcal{H})$ is contained in some $C \in \mathcal{C}$;
- every $C \in \mathcal{C}$ is almost independent in $\mathcal{H}$: $|\mathcal{H}[C]| = o(|\mathcal{H}|)$.

Container Lemma $\Rightarrow$ Transference A. If $p \gg p_{\mathcal{H}}$, then standard tail bounds and a union bound* give, a.a.s.,

$$\alpha(\mathcal{H}[V_p]) \leqslant \max_{C \in \mathcal{C}} |C \cap V_p| \leqslant p \cdot \max_{C \in \mathcal{C}} |C| + o(pN) = p\alpha(\mathcal{H}) + o(pN).$$

*This step is a little more subtle when $p = O(p_{\mathcal{H}} \log(1/p_{\mathcal{H}}))$.

The two approaches are incomparable

Transference B gives information about all subsets of V_p, not only independent sets. For instance, it can determine

$$\max\left\{|G| : G \subseteq G_{n,p}, |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3\right\},$$

provided that $p \gg n^{-1/2}$.

The two approaches are incomparable

Transference B gives information about all subsets of V_p, not only independent sets. For instance, it can determine

$$\max\left\{|G| : G \subseteq G_{n,p}, |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3\right\},$$

provided that $p \gg n^{-1/2}$.

Container Lemma implies counting statements. For example,

$$\#\{G \subseteq K_n : |G| = m,\ G \text{ is triangle-free}\} = \binom{(1/4 + o(1))n^2}{m},$$

provided that $m \gg n^{3/2}$.

The two approaches are incomparable

Transference B gives information about all subsets of V_p, not only independent sets. For instance, it can determine

$$\max\left\{|G| : G \subseteq G_{n,p}, |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3\right\},$$

provided that $p \gg n^{-1/2}$.

Container Lemma implies counting statements. For example,

$$\#\{G \subseteq K_n : |G| = m,\ G \text{ is triangle-free}\} = \binom{(1/4 + o(1))n^2}{m},$$

provided that $m \gg n^{3/2}$.

Motivating goal

Bridge the gap between Transference B and the Container Lemma.

The number of graphs with few triangles

Problem (Lower tails in $G(n,m)$)

For $G \sim G(n,m)$, determine the asymptotics of

$$\log \mathbb{P}\left(|\mathcal{T}_n[G]| \leqslant \frac{1}{2}\,\mathbb{E}|\mathcal{T}_n[G]|\right), \qquad m \gg n^{3/2}.$$

The number of graphs with few triangles

Problem (Lower tails in $G(n, m)$)

For $G \sim G(n, m)$, determine the asymptotics of

$$\log \mathbb{P}\left(|\mathcal{T}_n[G]| \leqslant \frac{1}{2}\,\mathbb{E}|\mathcal{T}_n[G]|\right), \qquad m \gg n^{3/2}.$$

This simply stated problem is still open in the range

$$n^{3/2} \ll m \leqslant n^{2-o(1)}.$$

The number of graphs with few triangles

Problem (Lower tails in $G(n, m)$)

For $G \sim G(n, m)$, determine the asymptotics of

$$\log \mathbb{P}\left(|\mathcal{T}_n[G]| \leqslant \frac{1}{2}\,\mathbb{E}|\mathcal{T}_n[G]|\right), \qquad m \gg n^{3/2}.$$

This simply stated problem is still open in the range

$$n^{3/2} \ll m \leqslant n^{2-o(1)}.$$

The sparse range. This is 'easy' when $m \ll n^{3/2}$. Beautiful (but partial) result of Jenssen, Perkins, Potukuchi, and Simkin for $m = \Theta(n^{3/2})$.

The number of graphs with few triangles

Problem (Lower tails in $G(n, m)$)

For $G \sim G(n, m)$, determine the asymptotics of

$$\log \mathbb{P}\left(|\mathcal{T}_n[G]| \leqslant \frac{1}{2}\, \mathbb{E}|\mathcal{T}_n[G]| \right), \qquad m \gg n^{3/2}.$$

This simply stated problem is still open in the range

$$n^{3/2} \ll m \leqslant n^{2-o(1)}.$$

The sparse range. This is 'easy' when $m \ll n^{3/2}$. Beautiful (but partial) result of Jenssen, Perkins, Potukuchi, and Simkin for $m = \Theta(n^{3/2})$.

The binomial model. When $G \sim G_{n,p}$, the work of Kozma and S. (2023) solves this problem for arbitrary uniform hypergraphs $\mathcal{H}$ and all $p \gg p_{\mathcal{H}}$.

A measure of dependency

Key Definition

Suppose that $R \subseteq V$ is a random set. Define

$$\mathrm{dep}(R, \mathcal{H}) := \sum_{A \in \mathcal{H}} \left| \mathbb{P}(A \subseteq R) - \prod_{v \in A} \mathbb{P}(v \in R) \right|.$$

A measure of dependency

Key Definition

Suppose that $R \subseteq V$ is a random set. Define

$$\mathrm{dep}(R, \mathcal{H}) := \sum_{A \in \mathcal{H}} \left| \mathbb{P}(A \subseteq R) - \prod_{v \in A} \mathbb{P}(v \in R) \right|.$$

Informally speaking, small $\mathrm{dep}(R, \mathcal{H})$ means that R is essentially independent along most edges of $\mathcal{H}$.

Our result

Theorem* (Kozma–S.–Shapira–Wagner, 2026+)

Suppose that $\mathcal{H}$ is an r-uniform hypergraph, $p \gg p_{\mathcal{H}}$, and $m = pN$. For **every** $\mathcal{P} \subseteq \binom{V}{m}$, there exist an exceptional family $\mathcal{P}_0 \subseteq \mathcal{P}$ and a partition

$$\mathcal{P} \setminus \mathcal{P}_0 = \bigcup_{i \in I} \mathcal{P}_i$$

such that

$$|\mathcal{P}_0| \leqslant e^{-\omega(m)} \cdot \binom{N}{m}, \qquad |I| = \exp(o(m)),$$

and, for most $i \in I$, if R is chosen uniformly from $\mathcal{P}_i$, then

$$\mathrm{dep}(R, \mathcal{H}) \ll p^r \cdot |\mathcal{H}|.$$

Our result

Theorem* (Kozma–S.–Shapira–Wagner, 2026+)

Suppose that $\mathcal{H}$ is an r-uniform hypergraph, $p \gg p_{\mathcal{H}}$, and $m = pN$. For **every** $\mathcal{P} \subseteq \binom{V}{m}$, there exist an exceptional family $\mathcal{P}_0 \subseteq \mathcal{P}$ and a partition

$$\mathcal{P} \setminus \mathcal{P}_0 = \bigcup_{i \in I} \mathcal{P}_i$$

such that

$$|\mathcal{P}_0| \leqslant e^{-\omega(m)} \cdot \binom{N}{m}, \qquad |I| = \exp(o(m)),$$

and, for most $i \in I$, if R is chosen uniformly from $\mathcal{P}_i$, then

$$\mathrm{dep}(R, \mathcal{H}) \ll p^r \cdot |\mathcal{H}|.$$

* The formal statement is a little more technical.

Example application

Let $p = m/\binom{n}{2}$, assume $p \gg p_{\mathcal{T}_n} = n^{-1/2}$, and set

$$\mathcal{P} \coloneqq \left\{ G \in \binom{E(K_n)}{m} : |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3 \right\}.$$

Example application

Let $p = m/\binom{n}{2}$, assume $p \gg p_{\mathcal{T}_n} = n^{-1/2}$, and set

$$\mathcal{P} := \left\{ G \in \binom{E(K_n)}{m} : |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3 \right\}.$$

Apply the theorem to obtain

$$\mathcal{P} = \mathcal{P}_0 \cup \bigcup_{i \in I} \mathcal{P}_i, \qquad |\mathcal{P}_0| \leqslant e^{-\omega(m)} \binom{\binom{n}{2}}{m}, \qquad |I| = \exp(o(m)).$$

Example application

Let $p = m/\binom{n}{2}$, assume $p \gg p_{\mathcal{T}_n} = n^{-1/2}$, and set

$$\mathcal{P} := \left\{ G \in \binom{E(K_n)}{m} : |\mathcal{T}_n[G]| \leqslant \frac{1}{2}\binom{n}{3}p^3 \right\}.$$

Apply the theorem to obtain

$$\mathcal{P} = \mathcal{P}_0 \cup \bigcup_{i \in I} \mathcal{P}_i, \qquad |\mathcal{P}_0| \leqslant e^{-\omega(m)} \binom{\binom{n}{2}}{m}, \qquad |I| = \exp(o(m)).$$

Hence

$$\log|\mathcal{P}| \leqslant \max\left\{ \log|\mathcal{P}_0|,\ \max_{i \in I} \log|\mathcal{P}_i| + o(m) \right\}.$$

Fix a non-exceptional part $\mathcal{P}_i$ and let G be uniformly random from $\mathcal{P}_i$. For $e \in E(K_n)$, put $q_e := \mathbb{P}(e \in G)$; then $\sum_e q_e = m$.

Example application

For the non-uniform binomial graph $G(n, \boldsymbol{q})$,

$$\mathbb{E}\#K_3(G(n, \boldsymbol{q})) = \sum_{ijk} q_{ij} q_{ik} q_{jk}.$$

Example application

For the non-uniform binomial graph $G(n, \boldsymbol{q})$,

$$\mathbb{E}\#K_3(G(n, \boldsymbol{q})) = \sum_{ijk} q_{ij} q_{ik} q_{jk}.$$

Since G belongs to a non-exceptional part,

$$\begin{aligned}\sum_{ijk} q_{ij} q_{ik} q_{jk} &\leqslant \sum_{ijk} \mathbb{P}(\{ij, ik, jk\} \subseteq G) + \mathrm{dep}(G, \mathcal{T}_n) \\ &\leqslant \left(\frac{1}{2} + o(1)\right) \binom{n}{3} p^3.\end{aligned}$$

Example application

For the non-uniform binomial graph $G(n, \boldsymbol{q})$,

$$\mathbb{E}\#K_3(G(n, \boldsymbol{q})) = \sum_{ijk} q_{ij} q_{ik} q_{jk}.$$

Since G belongs to a non-exceptional part,

$$\begin{aligned}\sum_{ijk} q_{ij} q_{ik} q_{jk} &\leqslant \sum_{ijk} \mathbb{P}(\{ij, ik, jk\} \subseteq G) + \mathrm{dep}(G, \mathcal{T}_n) \\ &\leqslant \left(\frac{1}{2} + o(1)\right) \binom{n}{3} p^3.\end{aligned}$$

Therefore, by subadditivity of entropy,

$$\begin{aligned}\log |\mathcal{P}_i| \leqslant \max \Bigg\{ \sum_{e \in E(K_n)} H(q_e) : \sum_e q_e = m, \\ \mathbb{E}\#K_3(G(n, \boldsymbol{q})) \leqslant \left(\frac{1}{2} + o(1)\right) \binom{n}{3} p^3 \Bigg\}.\end{aligned}$$

Here $H(q)$ denotes the entropy of a Bernoulli(q) random variable.


THANK YOU!
Questions?